

Selected Topics Of Living Systems HW2

Daniel Haim Breger, 316136944

Technion

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1 Linear Stability and Waves of Dry Active Polar Matter

The dynamics of dry polar active matter is given by

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot \left(-\gamma_\rho \nabla \frac{\delta F}{\delta \rho} + v_0 \rho \mathbf{p} \right) \quad (1)$$

$$\frac{\partial \mathbf{p}}{\partial t} + \lambda (\mathbf{p} \cdot \nabla) \mathbf{p} = -\gamma_p \frac{\delta F}{\delta \mathbf{p}} \quad (2)$$

We focus on the polar phase in 2D with $\rho = \rho_0 + \delta\rho(\mathbf{r})$ and $\mathbf{p} = \hat{\mathbf{x}} + \delta p(\mathbf{r})\hat{\mathbf{y}}$ where $|\delta\rho| \ll \rho_0$ and $|\delta p| \ll 1$ and consider the free energy

$$F = \int d^2r \left[\frac{\chi}{2} \left(\frac{\delta\rho}{\rho_0} \right)^2 - \frac{1}{2} h_\parallel p_\alpha p_\alpha + \frac{K}{2} \partial_\alpha p_\beta \partial_\alpha p_\beta - \omega_1 \frac{\delta\rho}{\rho_0} \nabla \cdot \mathbf{p} \right] \quad (3)$$

Here, we have used the summation convention. Note that $h_\parallel$ is a Lagrange multiplier to ensure that $p_\alpha p_\alpha = 1$. In class, we considered $\delta p(\mathbf{r}) = \delta p(y)$ and $\omega_1 > 0$. Perform a linear stability analysis and find the dispersion relation for (a) $\delta p = \delta p(y)$ and $\omega_1 < 0$ (b) general $\delta p(\mathbf{r})$ and $\omega_1 > 0$. In which case is the system unstable? Explain why. When is it stable - what is the velocity of propagating waves?

Before starting the specific cases, I'll rewrite the free energy using the known relations for the density and polarization.

$$p_\alpha p_\alpha = \mathbf{p} \cdot \mathbf{p} = 1 + (\delta p(\mathbf{r}))^2 \quad (4)$$

$$\partial_\alpha p_\beta \partial_\alpha p_\beta = (\partial_x \delta p(\mathbf{r}))^2 + (\partial_y \delta p(\mathbf{r}))^2 \quad (5)$$

$$\nabla \cdot \mathbf{p} = \partial_y \delta p(\mathbf{r}) \quad (6)$$

$$\frac{\delta\rho}{\rho_0} = \frac{\delta\rho(\mathbf{r})}{\rho_0} \quad (7)$$

now I can insert this into the free energy

$$F = \int d^2r \left[\frac{\chi}{2} \left(\frac{\delta\rho}{\rho_0} \right)^2 - \frac{1}{2} h_\parallel p_\alpha p_\alpha + \frac{K}{2} \partial_\alpha p_\beta \partial_\alpha p_\beta - \omega_1 \frac{\delta\rho}{\rho_0} \nabla \cdot \mathbf{p} \right] \quad (8)$$

$$= \int d^2r \left[\frac{\chi}{2} \left(\frac{\delta\rho(\mathbf{r})}{\rho_0} \right)^2 - \frac{1}{2} h_\parallel \left(1 + (\delta p(\mathbf{r}))^2 \right) + \frac{K}{2} (\partial_x \delta p(\mathbf{r}))^2 + (\partial_y \delta p(\mathbf{r}))^2 - \omega_1 \frac{\delta\rho(\mathbf{r})}{\rho_0} \partial_y \delta p(\mathbf{r}) \right] \quad (9)$$

next we find

$$\frac{\delta F}{\delta \rho} = \chi \frac{\delta\rho(\mathbf{r})}{\rho_0^2} - \frac{\omega_1}{\rho_0} \partial_y \delta p(\mathbf{r}) \quad (10)$$

to do the other variation I'll look at it term by term for $p_\alpha \rightarrow p_\alpha + \varepsilon \eta_\alpha$

$$\delta(p_\alpha p_\alpha) = p_\alpha^2 + 2\varepsilon p_\alpha \eta_\alpha + \varepsilon^2 \eta_\alpha^2 = 2p_\alpha \eta_\alpha \quad (11)$$

$$\delta(\partial_\alpha p_\beta \partial_\alpha p_\beta) = 2\partial_\alpha p_\beta \partial_\alpha (\delta p_\beta) \quad (12)$$

which needs an integration by parts

$$\int d^2r 2\partial_\alpha p_\beta \partial_\alpha (\delta p_\beta) = - \int d^2r 2\partial_\alpha \partial_\alpha p_\beta \delta p_\beta = - \int d^2r 2\nabla^2 p_\beta \delta p_\beta \quad (13)$$

with the boundary terms cancelling out because it's a variation. And finally

$$\delta \left(\frac{\delta \rho}{\rho_0} \nabla \cdot \mathbf{p} \right) = \frac{\delta \rho}{\rho_0} \partial_\alpha \delta p_\alpha \quad (14)$$

integrating by parts

$$\int d^2r \frac{\delta \rho}{\rho_0} \partial_\alpha \delta p_\alpha = - \int d^2r \partial_\alpha \left(\frac{\delta \rho}{\rho_0} \right) \delta p_\alpha \quad (15)$$

connecting everything back

$$\frac{\delta F}{\delta \mathbf{p}} = -h_\parallel \mathbf{p} - K \nabla^2 \mathbf{p} + \omega_1 \nabla \left(\frac{\delta \rho}{\rho_0} \right) \quad \frac{\delta F}{\delta \rho} = \chi \frac{\delta \rho(\mathbf{r})}{\rho_0^2} - \frac{\omega_1}{\rho_0} \partial_y \delta p(\mathbf{r}) \quad (16)$$

Looking at the dynamic equation for density

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot \left(-\gamma_\rho \nabla \frac{\delta F}{\delta \rho} + v_0 \rho \mathbf{p} \right) \quad (17)$$

$$= \gamma_\rho \nabla^2 \left(\frac{\delta F}{\delta \rho} \right) - \nabla \cdot (v_0 \rho \mathbf{p}) \quad (18)$$

inserting the density and assuming ρ_0 is constant in time

$$\frac{\partial \delta \rho(\mathbf{r})}{\partial t} = \gamma_\rho \nabla^2 \left(\frac{\delta F}{\delta \rho} \right) - \nabla \cdot (v_0 \rho \mathbf{p}) \quad (19)$$

where

$$\nabla \cdot (v_0 (\rho_0 + \delta \rho(\mathbf{r})) \mathbf{p}) \simeq v_0 (\partial_x \delta \rho(\mathbf{r}) + \rho_0 \partial_y \delta p(\mathbf{r})) \quad (20)$$

where we omitted the term quadratic in delta. Next I linearize the second equation

$$\frac{\partial \mathbf{p}}{\partial t} + \lambda (\mathbf{p} \cdot \nabla) \mathbf{p} = \frac{\partial \mathbf{p}}{\partial t} + \lambda ((\hat{\mathbf{x}} + \delta p(\mathbf{r}) \hat{\mathbf{y}}) \cdot \nabla) (\hat{\mathbf{x}} + \delta p(\mathbf{r}) \hat{\mathbf{y}}) \quad (21)$$

$$= \frac{\partial \mathbf{p}}{\partial t} + \lambda (\hat{\mathbf{x}} \cdot \nabla) \hat{\mathbf{x}} + \lambda (\delta p(\mathbf{r}) \hat{\mathbf{y}} \cdot \nabla) \hat{\mathbf{x}} + \lambda (\hat{\mathbf{x}} \cdot \nabla) \delta p(\mathbf{r}) \hat{\mathbf{y}} \quad (22)$$

$$= \frac{\partial \mathbf{p}}{\partial t} + \lambda \partial_x \hat{\mathbf{x}} + \lambda \delta p(\mathbf{r}) \partial_y \hat{\mathbf{x}} + \lambda \partial_x \delta p(\mathbf{r}) \hat{\mathbf{y}} \quad (23)$$

$$= \frac{\partial \mathbf{p}}{\partial t} + \lambda \partial_x \delta p(\mathbf{r}) \hat{\mathbf{y}} \quad (24)$$

and the right hand side

$$-\gamma_p \frac{\delta F}{\delta \mathbf{p}} = \gamma_p \left(h_\parallel \mathbf{p} + K \nabla^2 \mathbf{p} - \omega_1 \nabla \left(\frac{\delta \rho}{\rho_0} \right) \right) \quad (25)$$

therefore

$$\frac{\partial \mathbf{p}}{\partial t} + \lambda \partial_x \delta p(\mathbf{r}) \hat{\mathbf{y}} = \gamma_p \left(h_\parallel \mathbf{p} + K \nabla^2 \mathbf{p} - \omega_1 \nabla \left(\frac{\delta \rho}{\rho_0} \right) \right) \quad (26)$$

inserting the known expressions

$$\partial_t (\hat{\mathbf{x}} + \delta p(\mathbf{r})\hat{\mathbf{y}}) + \lambda \partial_x \delta p(\mathbf{r})\hat{\mathbf{y}} = \gamma_p \left(h_{\parallel} (\hat{\mathbf{x}} + \delta p(\mathbf{r})\hat{\mathbf{y}}) + K \nabla^2 (\hat{\mathbf{x}} + \delta p(\mathbf{r})\hat{\mathbf{y}}) - \omega_1 \nabla \left(\frac{\delta \rho}{\rho_0} \right) \right) \quad (27)$$

focusing on the elements in the $\hat{\mathbf{y}}$ direction and omitting the element depending on $h_{\parallel}$ since it only contributes to quadratic terms

$$\partial_t \delta p(\mathbf{r}) + \lambda \partial_x \delta p(\mathbf{r}) = \gamma_p \left(K \nabla^2 \delta p(\mathbf{r}) - \omega_1 \nabla \left(\frac{\delta \rho}{\rho_0} \right)_y \right) \quad (28)$$

Now we focus on each case separately. In case (a), $\delta p = \delta p(y)$ and $\omega_1 < 0$. In that case

$$\partial_t \delta p(y) = \gamma_p \left(K \partial_y^2 \delta p(y) - \omega_1 \nabla \left(\frac{\delta \rho}{\rho_0} \right)_y \right) \quad (29)$$

and

$$\frac{\partial \delta \rho(y)}{\partial t} = \gamma_\rho \left(\partial_y^2 \chi \frac{\delta \rho(y)}{\rho_0^2} - \frac{\omega_1}{\rho_0} \partial_y^3 \delta p(y) \right) - v_0 \rho_0 \partial_y \delta p(y) \quad (30)$$

These equations look awful to solve (and I was stuck here for a while) but since I know the solution describes a wave, I propose an ansatz of the form

$$\delta \rho = \rho_A e^{i(ky - \omega t)} \quad \delta p = P_A e^{i(ky - \omega t)} \quad (31)$$

which using fourier transform identities gives

$$\partial_t = -i\omega \quad \partial_y = ik \quad \partial_y^2 = -k^2 \quad \partial_y^3 = -ik^3 \quad (32)$$

inserting these gives

$$-i\omega P_A e^{i(ky - \omega t)} = \gamma_p \left(-K k^2 P_A e^{i(ky - \omega t)} - ik\omega_1 \frac{\rho_A}{\rho_0} e^{i(ky - \omega t)} \right) \quad (33)$$

$$-i\omega \rho_A e^{i(ky - \omega t)} = \gamma_\rho \left(-k^2 \chi \frac{e^{i(ky - \omega t)}}{\rho_0} + \frac{\omega_1}{\rho_0} ik^3 P_A e^{i(ky - \omega t)} \right) - v_0 \rho_0 ik P_A e^{i(ky - \omega t)} \quad (34)$$

simplifying

$$-i\omega P_A = \gamma_p \left(-K k^2 P_A - ik\omega_1 \frac{\rho_A}{\rho_0} \right) \quad (35)$$

$$-i\omega \rho_A = \gamma_\rho \left(-k^2 \chi \frac{\rho_A}{\rho_0} + \frac{\omega_1}{\rho_0} ik^3 P_A \right) - v_0 \rho_0 ik P_A \quad (36)$$

in matrix representation

$$\begin{pmatrix} ik\omega_1 \frac{\gamma_p}{\rho_0} & K k^2 \gamma_p - i\omega \\ k^2 \chi \frac{\gamma_\rho}{\rho_0} - i\omega & ikv_0 \rho_0 - ik^3 \omega_1 \frac{\gamma_\rho}{\rho_0} \end{pmatrix} \begin{pmatrix} \rho_A \\ P_A \end{pmatrix} = 0 \quad (37)$$

Now we find the determinant

$$\begin{aligned}
 \begin{vmatrix} ik\omega_1 \frac{\gamma_p}{\rho_0} & Kk^2\gamma_p - i\omega \\ k^2\chi \frac{\gamma_p}{\rho_0} - i\omega & ikv_0\rho_0 - ik^3\omega_1 \frac{\gamma_p}{\rho_0} \end{vmatrix} &= \left(ik\omega_1 \frac{\gamma_p}{\rho_0} \right) \left(ikv_0\rho_0 - ik^3\omega_1 \frac{\gamma_p}{\rho_0} \right) - (Kk^2\gamma_p - i\omega) \left(k^2\chi \frac{\gamma_p}{\rho_0} - i\omega \right) \quad (38) \\
 &= -k^2v_0\omega_1\gamma_p + k^4\omega_1^2 \frac{\gamma_p\gamma_p}{\rho_0} - Kk^4\chi \frac{\gamma_p\gamma_p}{\rho_0} + i\omega Kk^2\gamma_p + i\omega k^2\chi \frac{\gamma_p}{\rho_0} + \omega^2 \quad (39) \\
 &= \omega^2 + i\omega k^2 \left(K\gamma_p + \chi \frac{\gamma_p}{\rho_0} \right) + \left(k^4 \frac{\gamma_p\gamma_p}{\rho_0} (\omega_1^2 - K\chi) - k^2v_0\omega_1\gamma_p \right) \quad (40)
 \end{aligned}$$

so the dispersion relations are

$$\omega_{\pm}(k) = i \frac{-k^2 \left(K\gamma_p + \chi \frac{\gamma_p}{\rho_0} \right) \pm k \sqrt{\left(K\gamma_p + \chi \frac{\gamma_p}{\rho_0} \right)^2 + 4 \left(k^4 \frac{\gamma_p\gamma_p}{\rho_0} (\omega_1^2 - K\chi) - v_0\omega_1\gamma_p \right)}}{2} \quad (41)$$

next if $\omega_1 < 0$, the expression in the square root is positive